Field Extensions and Elliptic Curves

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Blockstream Research

research.blockstream.com

github.com/BlockstreamResearch/zk-book

Blockstream

Plan

1 Field Extensions
- A bit of intuition
- General Definition
- Polynomial Quotient Ring
- Finite Field Extensions

2 Algebraic Closure
- Definition

3 Elliptic Curve
- Definition
- Group Structure

Field Extensions

$\mathbb{Q}$ vs $\mathbb{R}$

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$\mathbb{R}$ is sort of “an extended version of $\mathbb{Q}$”.

What about $\mathbb{R}$?

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$$(x_1, y_1) \cdot (x_2, y_2) = (x_1 x_2 - y_1 y_2, x_1 y_2 + x_2 y_1).$$

A bit about complex numbers

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Conclusion + Question

$\mathbb{C}$ is sort of "an extended version of $\mathbb{R}$". Thus, we have

$$\mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}, \text{ where } \mathbb{Q}, \mathbb{R}, \mathbb{C} \text{ are fields}$$

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$\mathbb{C}/\mathbb{R}$ is a field extension. So is $\mathbb{R}/\mathbb{Q}$.

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Similarly, $\mathbb{Q}(i) := \{p + qi : p, q \in \mathbb{Q}\}$ is a field extension of $\mathbb{Q}$.

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Notice

Each element of $\mathbb{Q}(\sqrt{2}, i)$ is a linear combination of $\{1, \sqrt{2}, i, \sqrt{2}i\}$. This is usually called a **basis**. Moreover, to denote the dimensionality of $\mathbb{Q}(\sqrt{2}, i)$ over $\mathbb{Q}$, we write $[\mathbb{Q}(\sqrt{2}, i) : \mathbb{Q}] = 4$.

Real Polynomials modulo $T^2 + 1$

Definition... "Kinda"

Consider the set $\mathcal{P}$ — a set of polynomials $\mathbb{R}[T]$ modulo $p(T) := T^2 + 1$.

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$$T^2 + 2T + 4 = (T^2 + 1) \cdot 1 + (2T + 3)$$

So in $\mathcal{P}$, we have $T^2 + 2T + 4 = 2T + 3$.

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Arithmetic

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Indeed,

$$\frac{1+T}{2} \cdot (1 - T) = \frac{1}{2} \cdot (1 - T^2) = \frac{1}{2}\left(-(T^2 + 1) + 2\right) = 1 \text{ (in } \mathcal{P})$$

Hold on a minute...

Results

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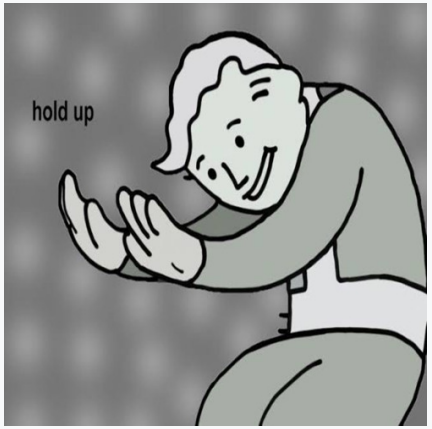
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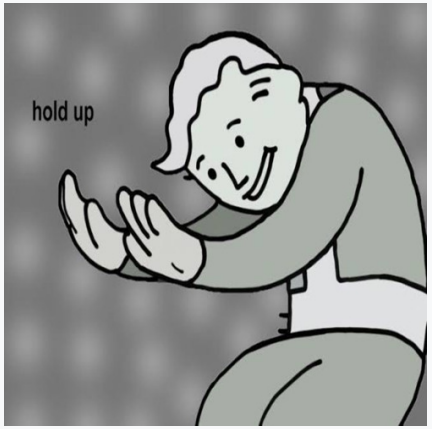
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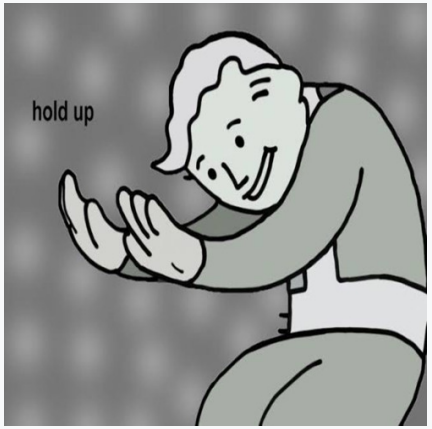


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Yes, any **irreducible** 2nd-degree polynomial $p(T)$ over $\mathbb{R}$ can be used. Typically, this is denoted as $\boxed{\mathbb{R}[T]/(p)}$.

Isomorphisms

Reminder

For two groups $(\mathbb{G}, +)$ and $(\mathbb{H}, \times)$ we defined homomorphism to be a function $\phi : \mathbb{G} \to \mathbb{H}$ such that

$$\phi(a + b) = \phi(a) \times \phi(b)$$

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- $\phi(1_{\mathbb{F}}) = 1_{\mathbb{K}}$

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But for now, $\cong$ means “exhibit the same structure”.

Key Theorems

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Let $\mathbb{F}$ be a field and $\mu(T)$ — irreducible polynomial over $\mathbb{F}$ (**reduction polynomial**). Consider a set of polynomials over $\mathbb{F}[T]$ modulo μ, formally denoted as $\mathbb{F}[T]/(\mu)$. Then, $\mathbb{F}[T]/(\mu)$ is a field.

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Theorem

Let $\mathbb{F}$ be a field and $\mu \in \mathbb{F}[T]$ is an irreducible polynomial of degree n and let $\mathbb{K} := \mathbb{F}[T]/(\mu)$. Let $\theta \in \mathbb{K}$ be the root of μ over $\mathbb{K}$. Then,

$$\mathbb{K} = \{c_0 + c_1\theta + \cdots + c_{n-1}\theta^{n-1} : c_0, \ldots, c_{n-1} \in \mathbb{F}\}$$

Coming back to previous examples

Example

Again, consider $\mathbb{Q}(\sqrt{2}) = \{p + q\sqrt{2} : p, q \in \mathbb{Q}\}$. Then,

$$\mathbb{Q}(\sqrt{2}) \cong \mathbb{Q}[T]/(T^2 - 2)$$

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Example

And $\mathbb{Q}(\sqrt{2}, i)$ is just a little bit more tricky. We cannot take $(T^2 - 2)(T^2 + 1)$ since it is reducible. Instead, notice that $\theta := \sqrt{2} + i$ is a root of the irreducible polynomial

$$p(T) := T^4 - 2T^2 + 9$$

So $\mathbb{Q}(\sqrt{2}, i) \cong \mathbb{Q}[T]/(T^4 - 2T^2 + 9)$.

Finite Field Extension

Definition

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Definition

Suppose p is prime and $m \geq 2$. Let $\mu \in \mathbb{F}_p[T]$ be an irreducible polynomial of degree m. Then, elements of $\mathbb{F}_{p^m}$ are polynomials in $\mathbb{F}_p[T]^{<m}$. In other words,

$$\mathbb{F}_{p^m} = \{c_0 + c_1 T + \cdots + c_{m-1} T^{m-1} : c_0, \ldots, c_{m-1} \in \mathbb{F}_p\},$$

where all operations are performed modulo μ.

Examples

It would be convenient to build $\mathbb{F}_{p^2}$ as $\mathbb{F}_p[i]/(i^2+1)$, but is it always possible? In other words, when $T^2 = -1$ has a solution in $\mathbb{F}_p$?

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- **Addition:** $(1+10i)+(18+15i) = 19+25i = 6i$.
- **Multiplication:** $(5+6i)(6+7i) = 30+71i+42i^2 = -12+71i = 7+14i$.

More Examples: Binary Extension Fields

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- **Inversion:** $(T^3 + T^2 + 1)^{-1} = T^2$ since $(T^3 + T^2 + 1) \cdot T^2$ mod $(T^4 + T + 1) = 1$.

More Examples: BN254

Example

Consider the **BN254 base field**, used in SNARKs:

$$p = \text{0x30644e72e131a029} \cdots \text{a8d3c208c16d87cfd47}$$

- Then, $\mathbb{F}_{p^2} := \mathbb{F}_p[u]/(u^2 + 1)$ since $p \equiv 3 \pmod 4$.

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- Finally, set $\mathbb{F}_{p^{12}} := \mathbb{F}_{p^6}[w]/(w^2 - v)$.

Equivalently, we can write:

$$\mathbb{F}_{p^{12}} := \mathbb{F}_p[w]/(w^{12} - 18w^6 + 82)$$

Algebraic Closure

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$\mathbb{R}$ is not algebraically closed since $T^2 + 1$ has no roots in $\mathbb{R}$. However, $\mathbb{C}$ is algebraically closed, which follows from the fundamental theorem of algebra. Since $\mathbb{C}$ is a field extension of $\mathbb{R}$, it is also an algebraic closure of $\mathbb{R}$. This is commonly denoted as $\overline{\mathbb{R}} = \mathbb{C}$.

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Definition

A field $\mathbb{K}$ is called an **algebraic closure** of $\mathbb{F}$ if $\mathbb{K}/\mathbb{F}$ is algebraically closed. This is denoted as $\overline{\mathbb{F}} = \mathbb{K}$.

Algebraic Closure for Finite Fields

Recall that we are cryptographers, not mathematicians. So we are interested in $\overline{\mathbb{F}}_p$. So I have two news to you:

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Clearly, $p(T)$ is a non-constant polynomial and has no roots in $\mathbb{F}$, since for any $f \in \mathbb{F}$, one has $p(f) = 1$. ■

So what?

But what form does the $\overline{\mathbb{F}}_p$ have? Well, it is a union of all $\mathbb{F}_{p^k}$ for $k \geq 1$. This is formally written as:

$$\boxed{\overline{\mathbb{F}}_p = \bigcup_{k \in \mathbb{N}} \mathbb{F}_{p^k}}$$

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Remark

But this definition is super counter-intuitive! So here how we usually interpret it. Suppose I tell you that polynomial $q(T)$ has a root in $\overline{\mathbb{F}}_p$. What that means is that there exists some extension $\mathbb{F}_{p^m}$ such that for some $\alpha \in \mathbb{F}_{p^m}$, $q(\alpha) = 0$. We do not know how large this m is, but we know that it exists. For that reason, $\overline{\mathbb{F}}_p$ is defined as an infinite union of all possible field extensions.

Elliptic Curve

Definition

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Suppose that $\mathbb{K}$ is a field. An **elliptic curve** E over $\mathbb{K}$ is defined as a set of points $(x, y) \in \mathbb{K}^2$:

$$y^2 = x^3 + ax + b,$$

called a **Short Weierstrass equation**, where $a, b \in \mathbb{K}$ and $4a^3 + 27b^2 \neq 0$. We denote $E/\mathbb{K}$ to denote the elliptic curve over field $\mathbb{K}$.

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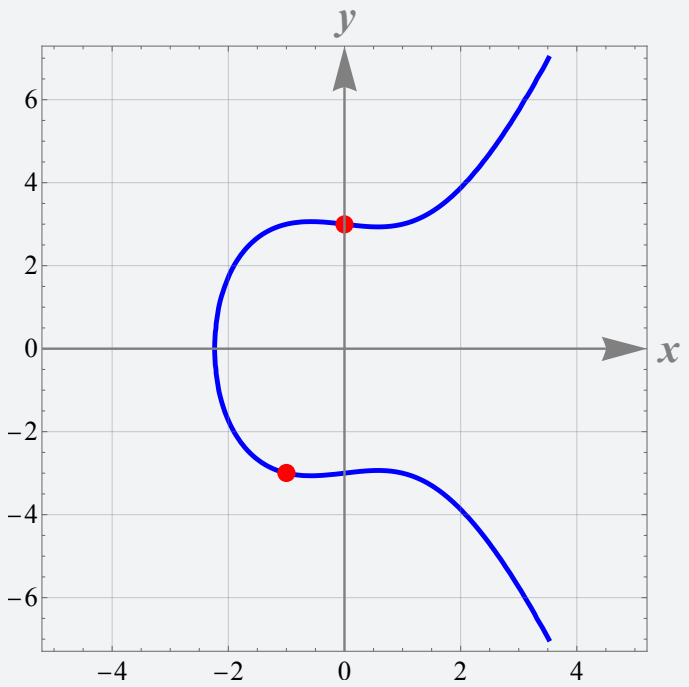
Definition

We say that $P = (x_P, y_P) \in \mathbb{A}^2(\mathbb{K})$ is the **affine representation** of the point on the elliptic curve $E/\mathbb{K}$ if it satisfies the equation $y_P^2 = x_P^3 + ax_P + b$.

Examples

Example

Consider $E/\mathbb{Q} : y^2 = x^3 - x + 9$. Valid affine points on $E/\mathbb{Q}$ are, for example, $P = (0, 3), Q = (-1, -3) \in \mathbb{A}^2(\mathbb{Q})$.



More Examples

Some more examples[1]:

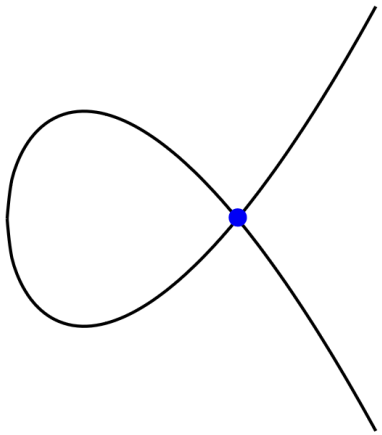


Figure 2.1: Singular curve $y^2 = x^3 - 3x + 2$ over $\mathbb{R}$.

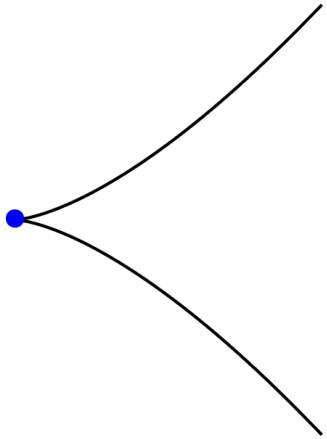


Figure 2.2: Singular curve $y^2 = x^3$ over $\mathbb{R}$.

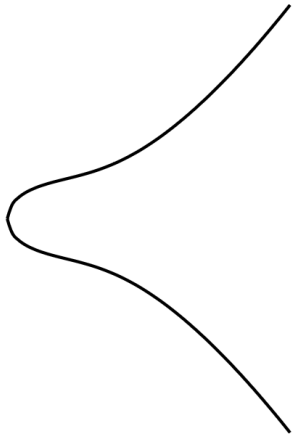


Figure 2.3: Smooth curve $y^2 = x^3 + x + 1$ over $\mathbb{R}$.

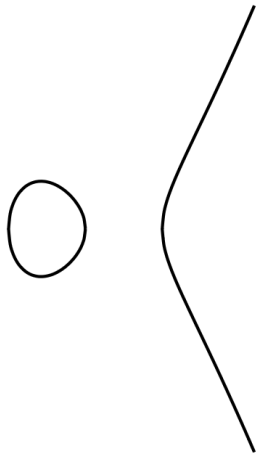


Figure 2.4: Smooth curve $y^2 = x^3 - x$ over $\mathbb{R}$.

[1] Figure taken from “Pairings for Beginners”

Real Elliptic Curves

But real elliptic curves are not that simple. Here how they look like[2]:

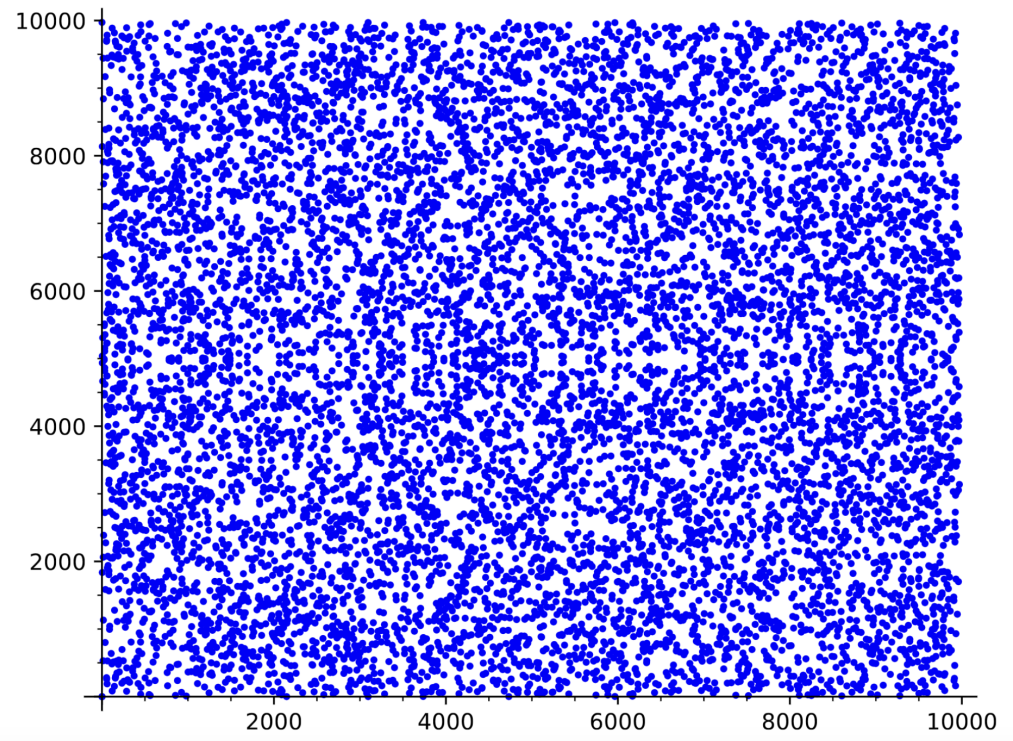


Figure: Curve $E/\mathbb{F}_{9973} : y^2 = x^3 - 2x + 1$ over the finite field

[2]Figure taken from “Moonmath”

Defining a Group Structure: A Few Words

Definition

The set of points on the curve, denoted as $E_{a,b}(\mathbb{K})$, is defined as:

$$E_{a,b}(\mathbb{K}) = \{(x, y) \in \mathbb{A}^2(\mathbb{K}) : y^2 = x^3 + ax + b\} \cup \{\mathcal{O}\},$$

where $\mathcal{O}$ is the so-called **point at infinity**.

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If $(x_P, y_P) \in E(\mathbb{K})$ then $(x_P, -y_P) \in E(\mathbb{K})$.

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Remark #1

If $(x_P, y_P) \in E(\mathbb{K})$ then $(x_P, -y_P) \in E(\mathbb{K})$.

Remark #2

Typically, $\mathbb{K} = \overline{\mathbb{F}}_p$: we do not concretize over which finite field we define the elliptic curve.

Defining a Group Structure: Chord Method

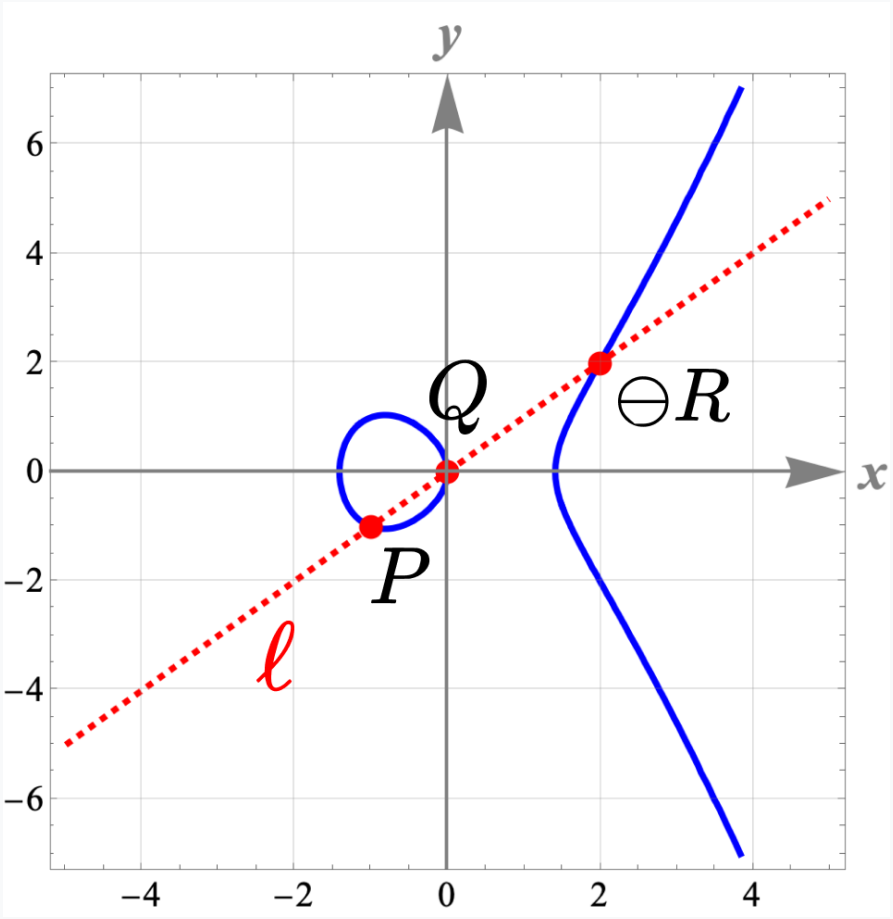


Figure: Chord method for adding two points

Defining a Group Structure: Tangent Method

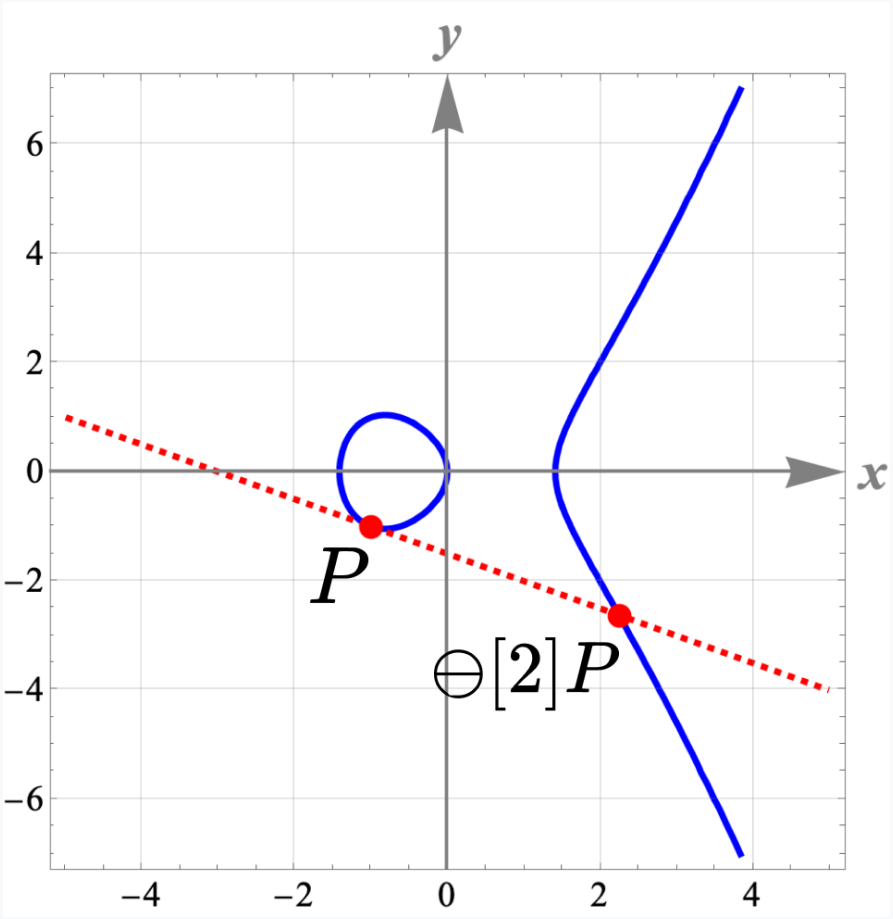


Figure: Tangent method for the point doubling

Idea of Derivation

Line equation through $P = (x_P, y_P), Q = (x_Q, y_Q)$:

$$\ell : y = \lambda(x - x_P) + y_P, \ \lambda = \frac{y_Q - y_P}{x_Q - x_P}$$

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Substituting y from the second equation to the first one, we get a cubic equation. Using Vieta's formula, one can derive

$$x_P + x_Q + x_R = \lambda^2$$

The rest is easy to finish.

Group Law

Definition

1. Point at infinity $\mathcal{O}$ is an identity element.

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$$x_R := \lambda^2 - x_P - x_Q, \quad y_R := \lambda(x_P - x_R) - y_P.$$

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3. If $x_P = x_Q$ and $y_P = y_Q$ (that is, $P = Q$), use the **tangent method**. Define the slope of the tangent at P as $\lambda := \frac{3x_P^2 + a}{2y_P}$ and set

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4. Otherwise, define $P \oplus Q := \mathcal{O}$.

One more Illustration

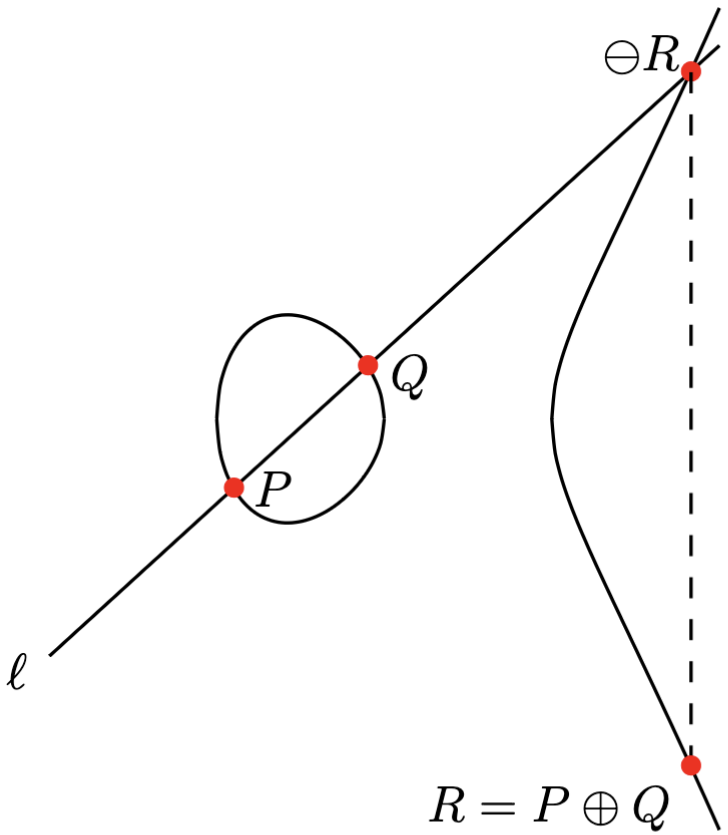


Figure 2.5: Elliptic curve addition.

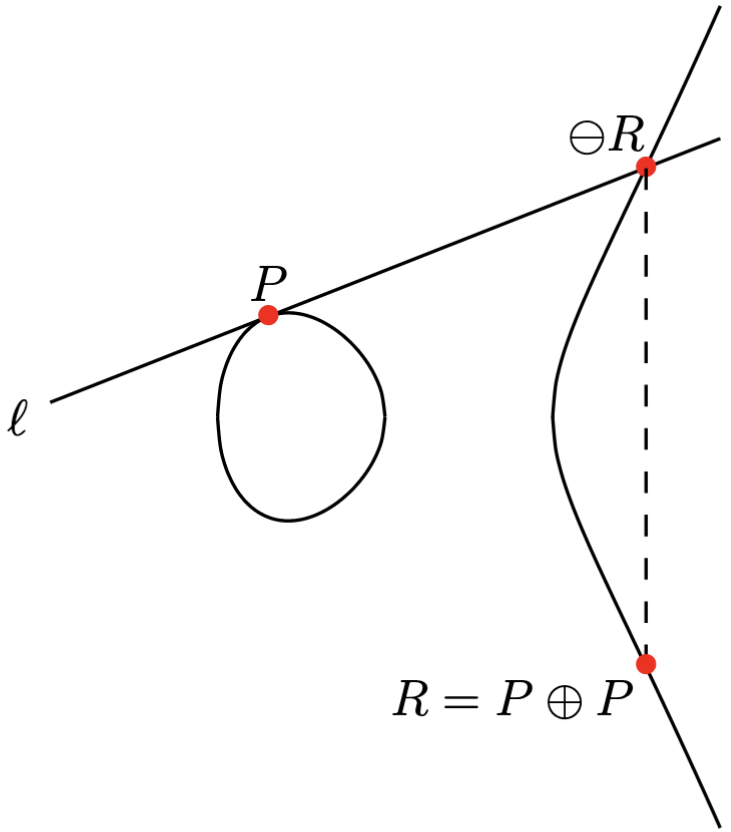


Figure 2.6: Elliptic curve doubling.

Example

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Consider $E/\mathbb{R} : y^2 = x^3 - 2x$.

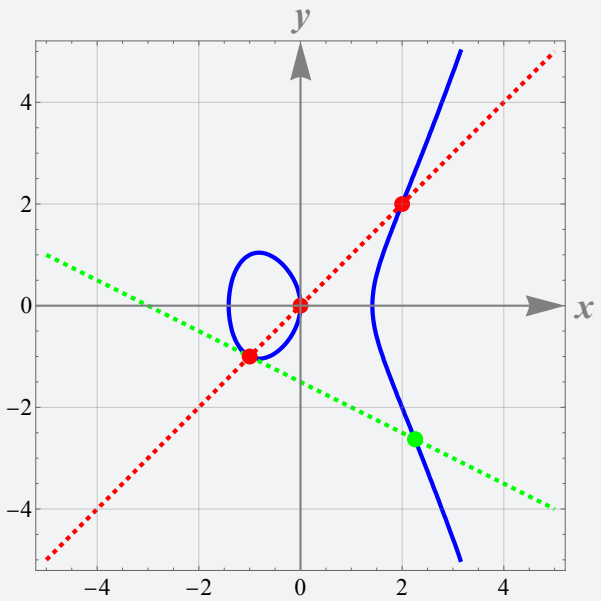
- **Addition:** $(-1,-1) \oplus (0,0) = (2,-2),\ (2,2) \oplus (-1,-1) = (0,0)$.

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- **Addition:** $(-1,-1) \oplus (0,0) = (2,-2),\ (2,2) \oplus (-1,-1) = (0,0).$
- **Doubling:** $[2](-1,-1) = \left(\frac{9}{4}, \frac{21}{8}\right).$



Hasse's Theorem

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Remark

In fact, $r = \#E(\mathbb{F}_{p^m})$ can be computed in polylogarithmic time $O(\log^\gamma(p^m))$ for some $\gamma > 1$, so the number of points can be computed efficiently even for fairly large primes p.

Discrete Logarithm

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Let $P \in E(\overline{\mathbb{F}}_p)$ and $\alpha \in \mathbb{Z}_r$. Define the scalar multiplication $[\alpha]P$ as adding P to itself $\alpha - 1$ times (also set $[0]P := \mathcal{O}$).

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Suppose E is cyclic, meaning, $\langle G \rangle = E$ for some $G \in E$. The **discrete logarithm problem** on E consists in the following: suppose $P = [\alpha]G$ for some $\alpha \in \mathbb{Z}_r$. Find α based on P.

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Remark

If r is a product of primes $p_1, p_2, \ldots, p_k$ such that $p_1 < p_2 < \cdots < p_k$, then the discrete logarithm problem can be solved in roughly $O(\sqrt{p_k})$: only the largest prime factor matters. For this reason, typically r is prime.

Thank you for your attention



research.blockstream.com

github.com/BlockstreamResearch/zk-book



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